

Performance Testing

Date: 30 September 2026 **Team ID:** 6ab3e57da238dc999eb3d44d **Project Name:** EduGenie – AI-Powered Educational Assistant

Maximum Marks: 5 Marks

Testing Overview

Field	Details
Testing Tool Used	Local Python compile check and application smoke testing; no dedicated JMeter/Locust/k6 report was supplied.
Type of Testing	Smoke / functional validation; formal load or stress testing is future work.
Target Modules / APIs	/, /health, /qa, /explain, /summarize, /quiz, /learn/recommendations
Test Environment	Local development environment; live Gemini performance depends on external service/network.
Test Date	30 September 2026

Test Scenarios

S.No	Scenario	Users	Duration	Expected Outcome	Observed / Evidence
1	Python source compilation	1	One run	No syntax errors	Pass – compileall returned success.
2	Open application home route	1	Manual	HTML page loads	Configured in source; runtime evidence not included in package.
3	Health endpoint	1	Manual	Returns status and AI live flag	Endpoint implemented in main.py.
4	Quiz response validation	1	Per request	Malformed JSON is rejected with useful error	Validation logic implemented in quiz_module.py.

Performance Metrics

Metric	Target	Actual Value	Status	Remarks
Average response time	< 2 sec	Not measured	Not assessed	AI latency varies with provider/network.
Maximum response time	< 5 sec	Not measured	Not assessed	Formal load testing not supplied.
Throughput	Project-specific baseline	Not measured	Not assessed	No load generator results supplied.
Error rate	< 1%	Not measured	Not assessed	Requires repeated/load testing.
CPU / Memory	< 80%	Not measured	Not assessed	Requires controlled runtime benchmark.

Conclusion: the supplied project supports functional/smoke validation, but numerical performance claims are deliberately left unreported until a controlled load test is run.